

1. Suppose X is a Hypergeometric random variable with parameters $n = 5$, $M = 20$, and $N = 30$.

1a. Write an exact expression for $P(X = 3)$ and calculate this value as a decimal number.

1b. Use a Binomial random variable to approximate the value in **1a** and calculate this approximate value.

(Hint: These two values should be close!)

2. Suppose X is a Binomial random variable with parameters $n = 7$ and $p = 1/2$. Suppose Y is a Hypergeometric random variable with parameters $n = 35$, $M = 20$, and $N = 200$. Calculate $\text{Var}(X)$ and $\text{Var}(Y)$.

3. Consider 300 green jelly beans and 200 red jelly beans in a jar. Suppose that Katie selects 8 jelly beans altogether (without replacement, and with all selections equally likely).

3a. What is the probability that she selects an equal number of green jelly beans and red jelly beans?

3b. Find the expected number of green jelly beans selected by Katie.

3c. Find the variance of the number of green jelly beans selected by Katie.

4. Suppose Matthias selects 7 balls altogether (without replacement, and with all selections equally likely) from a bucket that contains 25 green balls and 15 red balls.

Find the probability that Matthias selects *strictly more green balls* than Katie selects green jelly beans.

(It is safe to assume that their selections are independent.)

5. Consider 5 Biochemistry students, 3 Business Analytics students, 2 Mathematics students, 7 Physics students, and 8 Statistics students, who are taking an interdisciplinary Honors course. Suppose that 3 of the students are randomly selected to talk with a visiting alumna. While choosing the 3 students, all selections are equally likely. Let X denote the number of selected students who are studying Mathematics or Statistics. Find the probability mass function of X .